

**САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ
ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

Дисциплина: Бэк-энд разработка

Отчет

Лабораторная работа 4

Выполнил:

Сокович Степан

Группа К3344

Проверил:

Добряков Д. И.

Задание

провести предварительную настройку удалённого сервера (установка необходимых зависимостей),
провести развёртывание приложения на удалённом сервере, настроить nginx в режиме обратного прокси для обеспечения доступа к вашему приложению извне.

Ход работы:

1. Арендуем машину через compute в yandex cloud, воспользовавшись грантом

The screenshot displays the Yandex Cloud console interface for a virtual machine. The left sidebar contains navigation links: 'compute-vm-backend-lab' (Virtual Machine), 'Обзор' (Overview), 'Диски и хранилища' (Disks and Storage), 'Резервные копии' (Backups), 'Права доступа' (Permissions), 'Операции' (Operations), 'Мониторинг' (Monitoring), 'Серийная консоль' (Serial Console), and 'Последовательный порт' (Serial Port). Below these are links for documentation: 'Создать виртуальную машину' (Create Virtual Machine), 'Тарифы Compute Cloud' (Compute Cloud Tariffs), 'Настройка синхронизации времени NTP' (NTP Time Synchronization Settings), 'Подключиться к ВМ по SSH' (Connect to VM via SSH), 'Подключиться к ВМ через OS Login' (Connect to VM via OS Login), 'Графические ускорители GPU и vGPU' (GPU and vGPU Graphics Accelerators), and 'История изменений' (Change History).

The main panel shows the 'Обзор' (Overview) tab for the VM 'compute-vm-backend-lab'. The status is 'Running'. Key details include: Identifier: 'epd1lvjtc0o6pof91jje', Name: 'compute-vm-backend-lab', Creation Date: '26.05.2026, в 18:11', Internal FQDN: 'compute-vm-backend-lab.ru-central1.internal', and Availability Zone: 'ru-central1-b'.

The 'Доступ' (Access) section shows 'Доступ по OS Login' is 'Выключен' (Disabled). Below this, there are two expandable sections for connection instructions. The first, 'Подключиться с помощью SSH-клиента' (Connect via SSH client), provides the command: `ssh -l steven 158.160.26.87`. The second, 'Подключиться с помощью CLI Yandex Cloud' (Connect via Yandex Cloud CLI), is currently collapsed.

The 'Ресурсы' (Resources) section lists the VM's specifications: Platform: 'standard-v4a', Guaranteed vCPU share: '100%', vCPU: '2', RAM: '2 GB', and Disk Space: '0 B'.

At the bottom, there is a section for 'Приложение Marketplace' (Marketplace Application).

2. Подключимся к машине

```
undercat@undercat-Aspire-A315-59:~$ ssh -l steven 158.160.26.87
The authenticity of host '158.160.26.87 (158.160.26.87)' can't be established.
ED25519 key fingerprint is SHA256:dNMLnupN91ayC1GaW+ZHfeoANKPQ2KyrpspjklhTgI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '158.160.26.87' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-117-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue May 26 15:14:33 UTC 2026

System load:  0.33           Processes:            103
Usage of /:   9.0% of 23.57GB Users logged in:          0
Memory usage: 8%           IPv4 address for eth0: 10.129.0.27
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

steven@compute-vm-backend-lab:~$ ls
steven@compute-vm-backend-lab:~$
```

3. Установим зависимости.

```
steven@compute-vm-backend-lab:~$ ls
steven@compute-vm-backend-lab:~$ sudo apt install -y git nginx ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.9).
curl set to manually installed.
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
```

4. Установим докер

```
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
steven@compute-vm-backend-lab:~$ curl -fsSL https://get.docker.com | sudo sh
# Executing docker install script, commit: 2687d91ddeb3bd6a6ae37a90947761efdee87030
+ sh -c apt-get -qq update >/dev/null
+ sh -c DEBIAN_FRONTEND=noninteractive apt-get -y -qq install ca-certificates curl >/dev/null
+ sh -c install -m 0755 -d /etc/apt/keyrings
+ sh -c curl -fsSL "https://download.docker.com/linux/ubuntu/gpg" -o /etc/apt/keyrings/docker.asc
+ sh -c chmod a+r /etc/apt/keyrings/docker.asc
+ sh -c echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu noble stable" > /etc/apt/sources.list.d/docker.list
+ sh -c apt-get -qq update >/dev/null
+ sh -c DEBIAN_FRONTEND=noninteractive apt-get -y -qq install docker-ce docker-ce-cli containerd.io docker-compose-plugin docker-ce-rootless-extras docker-buildx-plugin docker-model
plugin >/dev/null
```

5. Проверим скачивание зависимостей, склонируем на вм наш репо

```
steven@compute-vm-backend-lab:~$ docker --version
Docker version 29.5.2, build 79eb04c
steven@compute-vm-backend-lab:~$ docker compose version
Docker Compose version v5.1.4
steven@compute-vm-backend-lab:~$ nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
steven@compute-vm-backend-lab:~$ git clone https://github.com/MrUndercat/ITMO-ACS-Backend-2026-C.git
Cloning into 'ITMO-ACS-Backend-2026-C'...
remote: Enumerating objects: 417, done.
remote: Counting objects: 100% (88/88), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 417 (delta 70), reused 56 (delta 56), pack-reused 329 (from 1)
Receiving objects: 100% (417/417), 1.76 MiB | 6.26 MiB/s, done.
Resolving deltas: 100% (177/177), done.
steven@compute-vm-backend-lab:~$
```

6. Соберем и запустим все приложения и инфра

```
[+] up 32/32nager] resolving provenance for metadata file
✓ Image postgres:16-alpine Pulled
✓ Image bitnami/legacy/kafka:latest Pulled
✓ Image hw5-rent-manager Built
✓ Image hw5-communication-manager Built
✓ Image hw5-user-manager Built
✓ Image hw5-gateway Built
✓ Network hw5_default Created
✓ Volume hw5_pg_user Created
✓ Volume hw5_pg_rent Created
✓ Volume hw5_pg_comm Created
✓ Container hw5-postgres-rent-1 Healthy
✓ Container hw5-postgres-user-1 Healthy
✓ Container hw5-postgres-comm-1 Healthy
✓ Container hw5-kafka-1 Started
✓ Container hw5-user-manager-1 Started
✓ Container hw5-rent-manager-1 Started
✓ Container hw5-communication-manager-1 Started
✓ Container hw5-gateway-1 Started
```

7. Настроим корневой локейшн в конфиге nginx (реверс-прокси).
Остальные локейшны не настраиваем, т.к. Роутинг на них -
ответственность gateway.

```
undercat@undercat-Aspire-A315-59: ~  
steven@compute-vm-backend-lab: ~/ITMO-ACS-Backend-2026-C/БР2.2/Соковы...  
GNU nano 7.2 /etc/nginx/sites-available/hw5  
server {  
    listen 80;  
    server_name 158.160.26.87;  
  
    location / {  
        proxy_pass http://127.0.0.1:3000;  
        proxy_http_version 1.1;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
    }  
}
```

8. Перезапустим балансер с новым конфигом.

```
steven@compute-vm-backend-lab: ~/ITMO-ACS-Backend-2026-C/БР2.2/Соковыч Степан/homeworks/hw5$ sudo nano /etc/nginx/sites-available/hw5  
steven@compute-vm-backend-lab: ~/ITMO-ACS-Backend-2026-C/БР2.2/Соковыч Степан/homeworks/hw5$ sudo ln -s /etc/nginx/sites-available/hw5 /etc/nginx/sites-enabled/  
steven@compute-vm-backend-lab: ~/ITMO-ACS-Backend-2026-C/БР2.2/Соковыч Степан/homeworks/hw5$ sudo rm -f /etc/nginx/sites-enabled/default  
steven@compute-vm-backend-lab: ~/ITMO-ACS-Backend-2026-C/БР2.2/Соковыч Степан/homeworks/hw5$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
steven@compute-vm-backend-lab: ~/ITMO-ACS-Backend-2026-C/БР2.2/Соковыч Степан/homeworks/hw5$ sudo systemctl reload nginx
```

9. Проверим работу прокси с локальной машины

```
undercat@undercat-Aspire-A315-59:~$ curl --no-proxy "*" http://158.160.26.87  
{  
  "service": "API-шлюз аренды жилья (ЛР2)",  
  "services": {  
    "users": "http://user-manager:3001",  
    "rent": "http://rent-manager:3002",  
    "communication": "http://communication-manager:3003"  
  }  
}  
undercat@undercat-Aspire-A315-59:~$
```